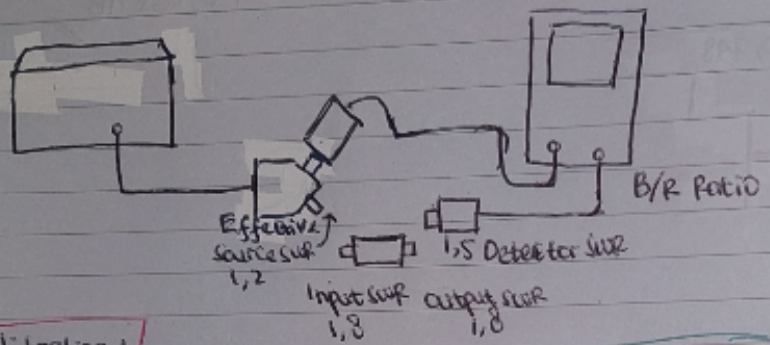
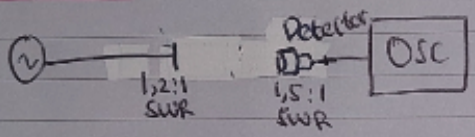
No
Date

Nama : Ridho Shofwan Rasyid
NIM : 22133102 7
Kelas : 2A - TC

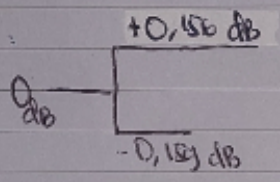
Contoh Soal Pengukuran High Frequency
Analyze the system to determine the uncertainty in measuring a 10 dB pad,
Assume no instrument error for frequency response error:



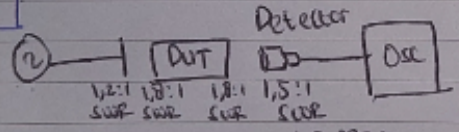
Calibration :



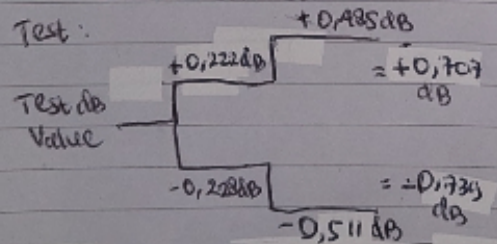
Calibration :



Test



Test :



Perhitungan

$$P = \frac{VSWR - 1}{VSWR + 1}$$

$$P_S = \frac{1,2 - 1}{1,2 + 1} = \frac{0,2}{2,2} = 0,0909$$

$$P_D = \frac{1,5 - 1}{1,5 + 1} = \frac{0,5}{2,5} = 0,2$$

$P_S \cdot P_d = 0,0909 \times 0,2 = 0,01818$

(+) $20 \log (1 + 0,01818) = +0,156$ dB

(−) $20 \log (1 - 0,01818) = -0,159$ dB

$$P_2 = P_1 = \frac{1,8 - 1}{1,8 + 1} = \frac{0,8}{2,8} = 0,2857$$

$P_S \cdot P_1 = 0,0909 \times 0,2857 = 0,02597$

$P_2 \cdot P_D = 0,2 \times 0,2857 = 0,05714$

(+) $20 \log (1 + 0,02597) = +0,222$ dB

(−) $20 \log (1 - 0,02597) = -0,228$ dB

(+) $20 \log (1 + 0,05714) = +0,482$ dB

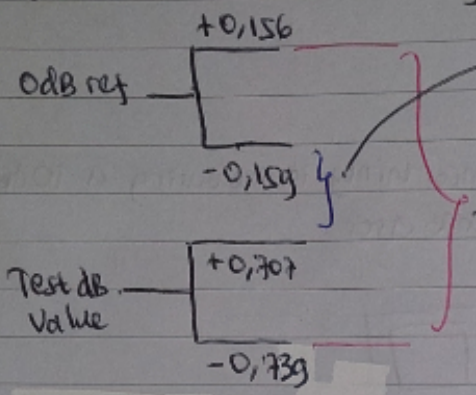
(−) $20 \log (1 - 0,05714) = -0,511$ dB

JOYKO 56 Lines, 6 mm

Ridho SR _ 221331027

No
Date

Total Mismatch Uncertainty



Min. Measured = -0,159 - (+0,707)
= -0,548 dB //

Max. Measured = +0,156 - (-0,739)
= +0,895 dB //

OYKO 36 Lines, 8 mm